

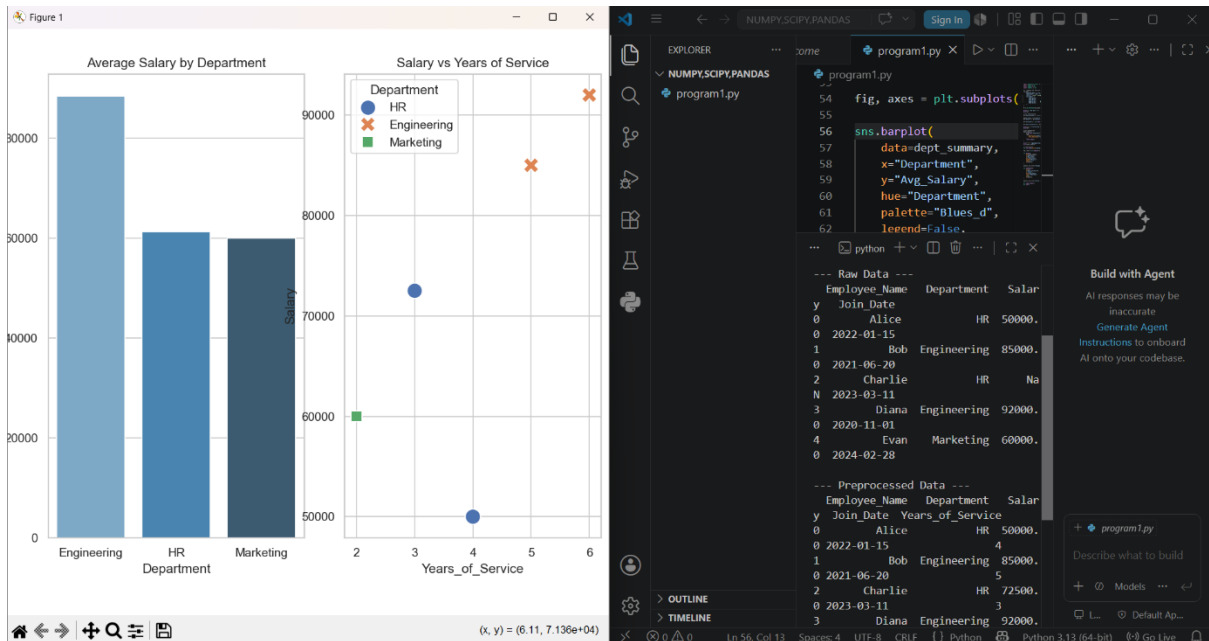
PRACTICE EXPERIMENTS

DSA0404-FDS

Name: K. L. Khaviya varshini

Reg. No.: 192472381

PROGRAM 1:



PROGRAM 2:

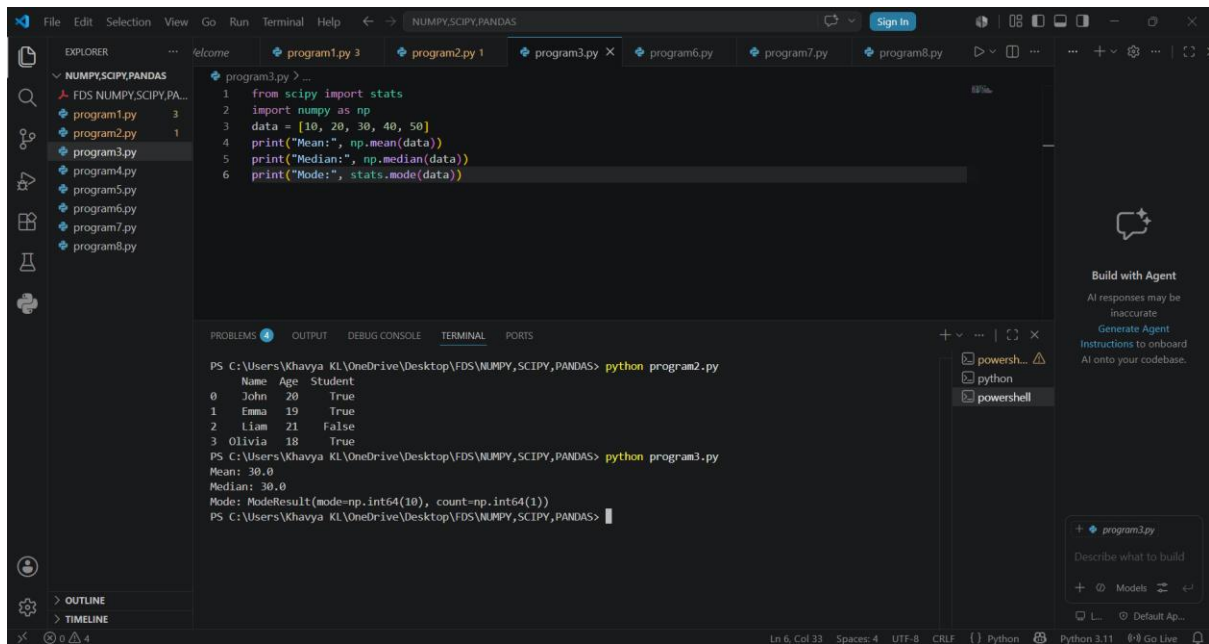
The screenshot displays a Jupyter Notebook interface with a Python script in a cell. The script creates a DataFrame with columns 'Name', 'Age', and 'Student'. The output of the script is shown in the terminal window at the bottom.

```
1 import pandas as pd
2 df = pd.DataFrame()
3 df['Name'] = ['John', 'Emma', 'Liam', 'Olivia']
4 df['Age'] = [20, 19, 21, 18]
5 df['Student'] = [True, True, False, True]
6 print(df)
```

Terminal Output:

```
PS C:\Users\Khaviya KL\OneDrive\Desktop\FDS\NUMPY, SCIPY, PANDAS> python program2.py
   Name  Age  Student
0  John   20     True
1  Emma   19     True
2  Liam   21    False
3  Olivia  18     True
PS C:\Users\Khaviya KL\OneDrive\Desktop\FDS\NUMPY, SCIPY, PANDAS>
```

PROGRAM 3:

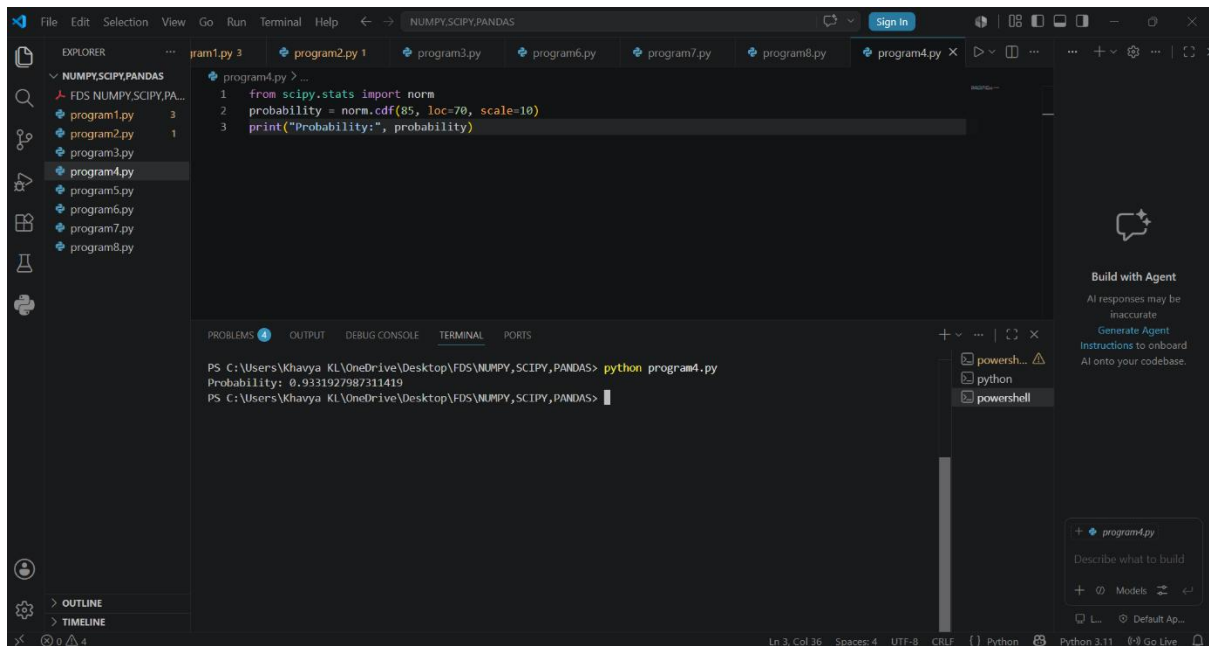


```
File Edit Selection View Go Run Terminal Help Numpy,Scipy,Pandas
EXPLORER welcome program1.py 3 program2.py 1 program3.py X program6.py program7.py program8.py
Numpy,Scipy,Pandas
FDS Numpy,Scipy,Pandas
program1.py 3
program2.py 1
program3.py
program4.py
program5.py
program6.py
program7.py
program8.py
program3.py > ...
1 from scipy import stats
2 import numpy as np
3 data = [10, 20, 30, 40, 50]
4 print("Mean:", np.mean(data))
5 print("Median:", np.median(data))
6 print("Mode:", stats.mode(data))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\Numpy,Scipy,Pandas> python program2.py
Name: Age Student
0 John 20 True
1 Emma 19 True
2 Liam 21 False
3 Olivia 18 True
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\Numpy,Scipy,Pandas> python program3.py
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\Numpy,Scipy,Pandas>

+ program3.py
Describe what to build
+ Models
+ Default Ap...
```

PROGRAM 4:

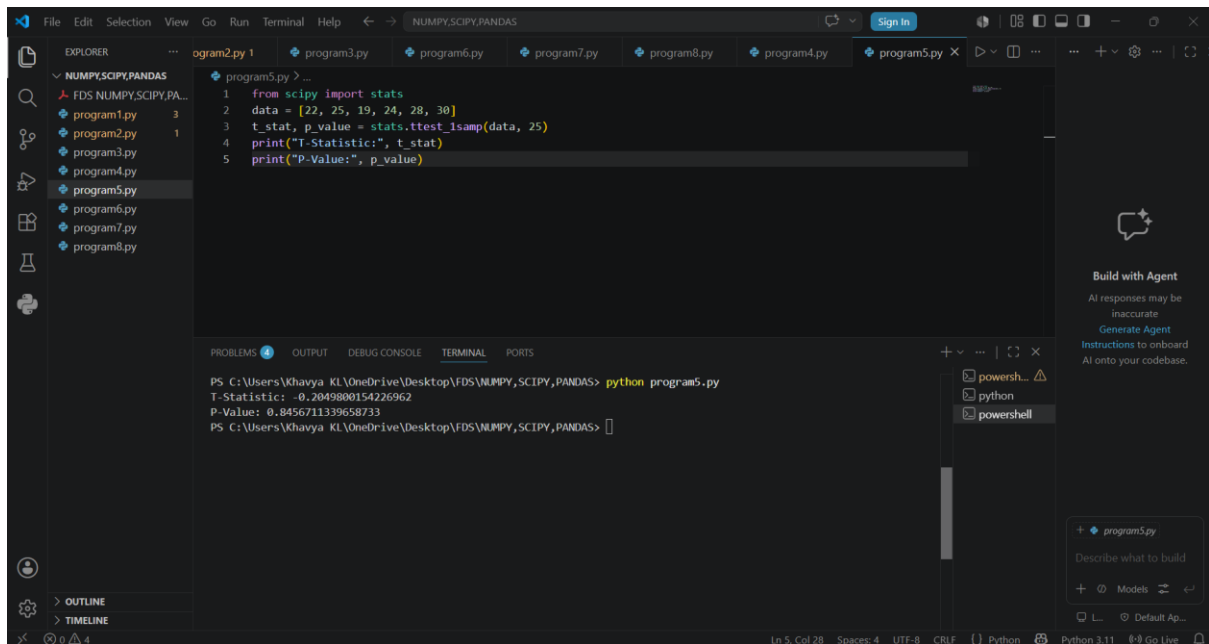


```
File Edit Selection View Go Run Terminal Help Numpy,Scipy,Pandas
EXPLORER welcome program1.py 3 program2.py 1 program3.py program6.py program7.py program8.py
Numpy,Scipy,Pandas
FDS Numpy,Scipy,Pandas
program1.py 3
program2.py 1
program3.py
program4.py
program5.py
program6.py
program7.py
program8.py
program4.py > ...
1 from scipy.stats import norm
2 probability = norm.cdf(85, loc=70, scale=10)
3 print("Probability:", probability)

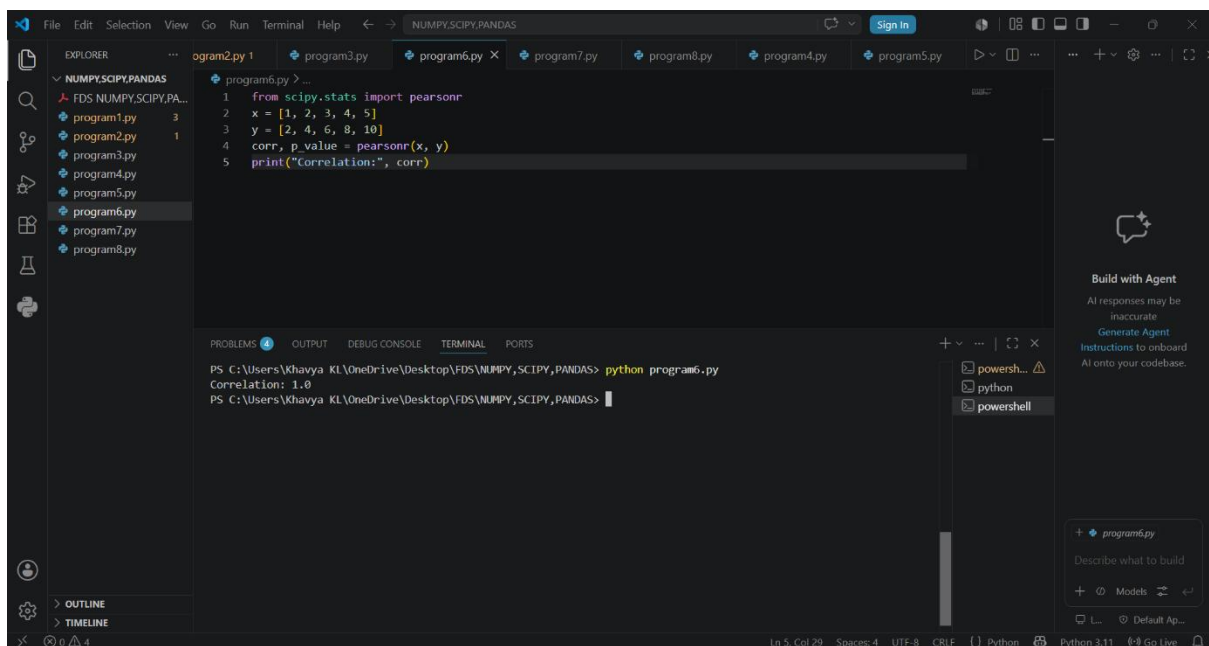
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\Numpy,Scipy,Pandas> python program4.py
Probability: 0.9331927987311419
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\Numpy,Scipy,Pandas>

+ program4.py
Describe what to build
+ Models
+ Default Ap...
```

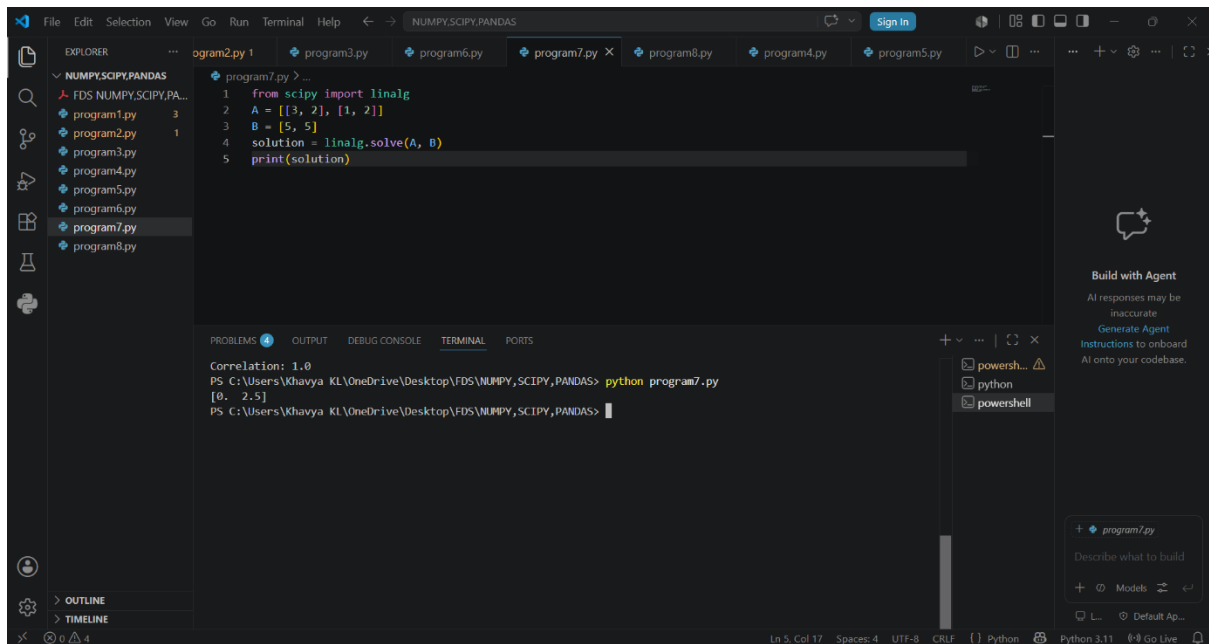
PROGRAM 5:



PROGRAM 6:



PROGRAM 7:



The screenshot shows the Visual Studio Code interface with the file explorer on the left displaying a project named 'NUMPY, SCIPY, PANDAS'. The file 'program7.py' is selected and open in the editor. The code in the editor is as follows:

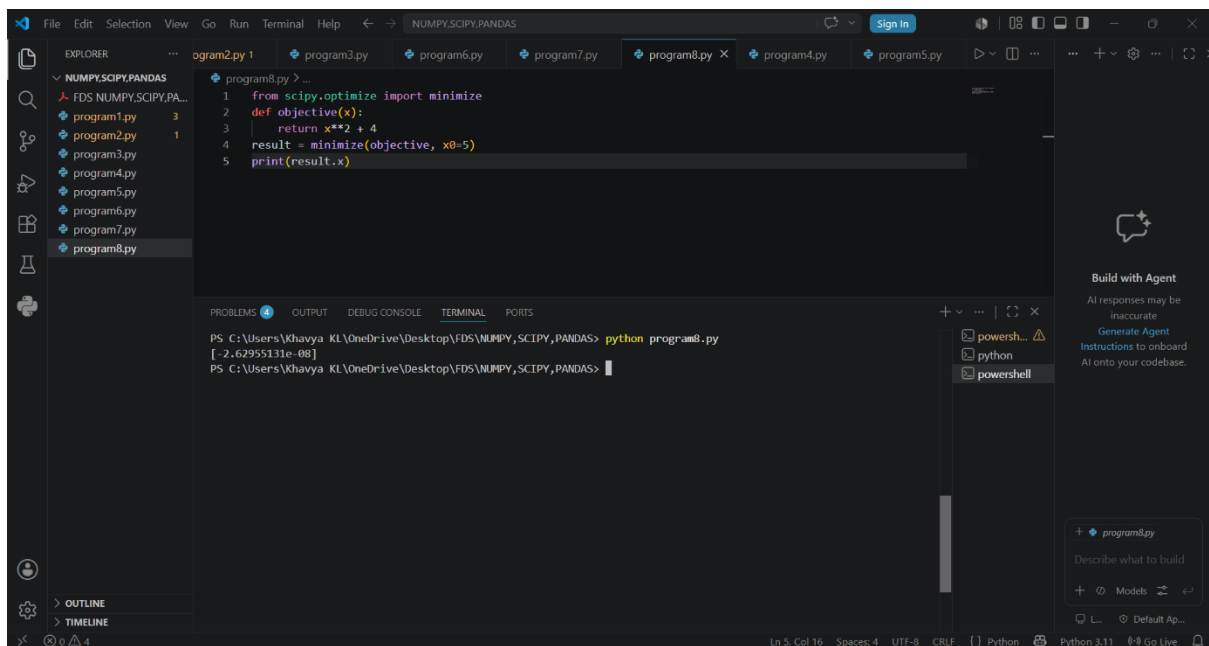
```
1 from scipy import linalg
2 A = [[3, 2], [1, 2]]
3 B = [5, 5]
4 solution = linalg.solve(A, B)
5 print(solution)
```

The terminal at the bottom shows the command `python program7.py` being executed, resulting in the output:

```
Correlation: 1.0
[0. 2.5]
```

The status bar at the bottom indicates the file is at line 5, column 17, using UTF-8 encoding, and the Python interpreter is set to Python 3.11.

PROGRAM 8:



The screenshot shows the Visual Studio Code interface with the file explorer on the left displaying the same project. The file 'program8.py' is selected and open in the editor. The code in the editor is as follows:

```
1 from scipy.optimize import minimize
2 def objective(x):
3     return x**2 + 4
4 result = minimize(objective, x0=5)
5 print(result.x)
```

The terminal at the bottom shows the command `python program8.py` being executed, resulting in the output:

```
[-2.62955131e-08]
```

The status bar at the bottom indicates the file is at line 5, column 16, using UTF-8 encoding, and the Python interpreter is set to Python 3.11.

The screenshot shows the VS Code interface with the following details:

- Explorer:** A file explorer on the left showing a project structure with files like `program1.py` through `program9.py`.
- Main Editor:** Displays the code for `program9.py`, which imports `numpy`, `matplotlib.pyplot`, `sklearn`, and `pandas` to load and preprocess the Iris dataset. The code includes `train_test_split` and `StandardScaler`.
- Terminal:** The bottom pane shows the terminal output of running the script, confirming the accuracy is 1.0.
- Right Sidebar:** Shows the 'Build with Agent' button and a 'powerhell' terminal window.

[illegible]